Lab 5B Doc

1. The young modulus is the slope of the linear elastic region. It is calculated as the change in stress divided by the change in strain

2. Endpoints:
Starting: (0,0)
A,B: (0.01,44)
C: (0.06, 44)
D: (0.18, 60)
E: (.27, 51)

3. Variables Needed:

strain

O_x
O_y

B_x
B_y
slope1

C_x
C_y
slope2

D_x
D_y
slope3

E_x
E_y
slope4

stress

4. Steps:

1. Ask the user for a strain value and make sure its a float
2. Enter the values of the endpoints into the variables B_x, B_y, C_x, C_y, etc
3. Find the slope between 0 to B_x and store it in slope1
4. Find the slope between B_x to C_x and store it in slope2
5. Find the slope between C_x and D_x and store it in slope3
6. Find the slope between D_x and E_x and store it in slope4
7. Write if elif statements to determine which segment the strain falls under
8. Use the slope to calculate the stress
9. Store the answer in the stress variable
10. Display the stress variable

5. Test Cases:

Typical:
Input: 0.001 | Output: 4.4
Input: 0.03 | Output: 44
Input: 0.12 | Output: 52
Input: 0.23 | Output: 55

Edges:
Input: -0.1 | Output: Error message
Input: 0 | Output: 0
Input: 0.01 | Output: 44
Input: 0.06 | Output: 44
Input: 0.18 | Output: 60
Input: 0.27 | Output: 51
Input: 0.28 | Output: Error message